

QUESTION:

Design a “farm animals” java application with the details of animals like cow, pig, horse.

Consider the following details like where they stay, what they eat, the sound they make by using classes and objects.

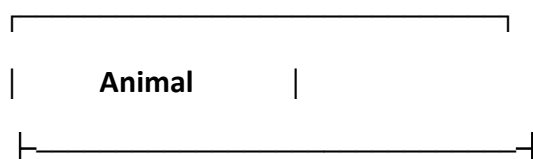
Steps for Execution

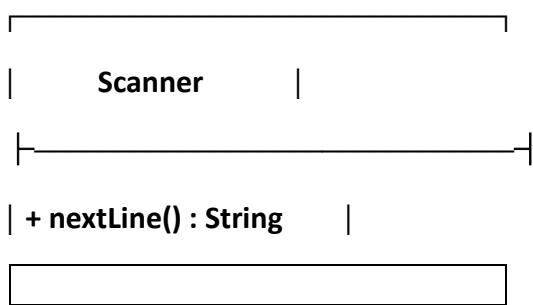
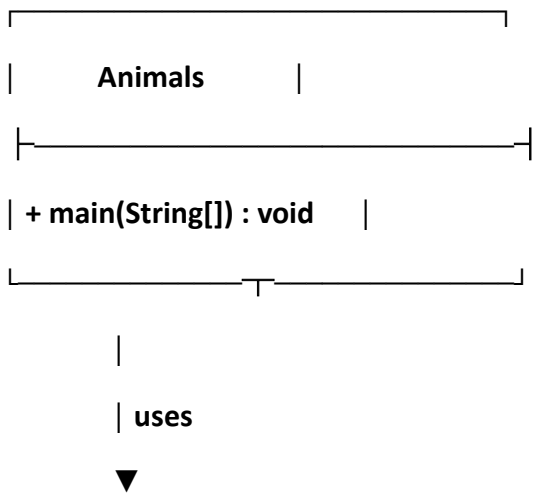
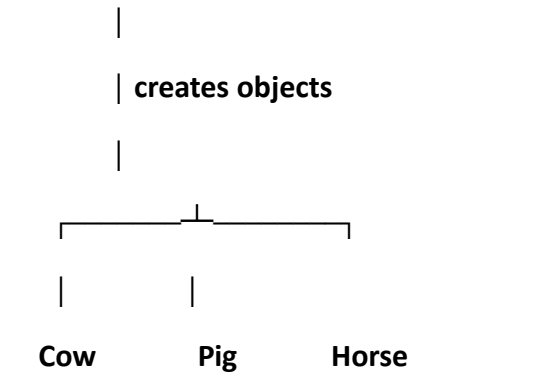
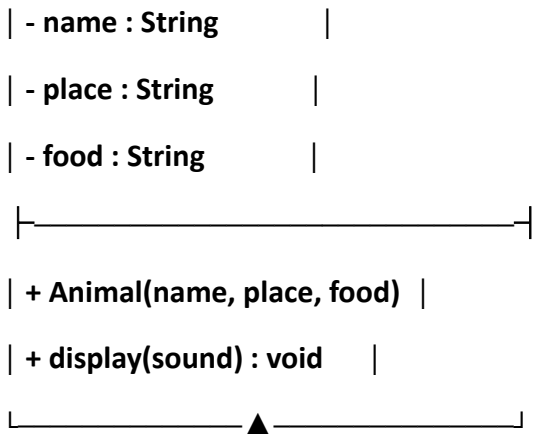
- 1.Open IntelliJ IDEA / Eclipse / VS Code.
- 2.Create a Java project.
- 3.Create a package named javaprograms.
- 4.Create a class named Animals.
- 5.Enter the given code and save it.
- 6.Run the Animals class.
- 7.Enter the place and food for cow, pig, and horse.
- 8.The program creates objects for all three animals.
- 9.Display their place, food, and sound.

Algorithm

- 1.Start.
- 2.Create a Scanner object for input.
- 3.Read the place and food of the cow.
- 4.Read the place and food of the pig.
- 5.Read the place and food of the horse.
- 6.Create Animal objects for cow, pig, and horse.
- 7.Call the display() method for each animal with its respective sound.
- 8.Display the animal's name, place, food, and sound.
- 9.Stop.

UML Class Diagram





CODE:

```
package javaprograms;
import java.util.Scanner;

class Animal {
    String name, place, food;

    public Animal(String name, String place, String food) {
        this.name = name;
        this.place = place;
        this.food = food;
    }

    void display(String sound) {
        System.out.println(name + " stays in " + place +
            ", eats " + food +
            ", makes " + sound + " sound");
    }
}

public class Animals {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter cow place: ");
        String cp = sc.nextLine();
        System.out.print("Enter cow food: ");
        String cf = sc.nextLine();

        System.out.print("Enter pig place: ");
        String pp = sc.nextLine();
        System.out.print("Enter pig food: ");
        String pf = sc.nextLine();

        System.out.print("Enter horse place: ");
        String hp = sc.nextLine();
        System.out.print("Enter horse food: ");
        String hf = sc.nextLine();

        Animal cow = new Animal("Cow", cp, cf);
```

```
Animal pig = new Animal("Pig", pp, pf);
Animal horse = new Animal("Horse", hp, hf);

cow.display("Moo");
pig.display("Oink");
horse.display("Neigh");
}
}
```

OUTPUT:

Enter cow place: Shed
Enter cow food: Grass
Enter pig place: Pen
Enter pig food: Grains
Enter horse place: Stable
Enter horse food: Grass

Cow stays in Shed, eats Grass, makes Moo sound
Pig stays in Pen, eats Grains, makes Oink sound
Horse stays in Stable, eats Grass, makes Neigh sound